Flow Debt Indicator

A6 – When one item finishes sooner, another one is paying for it

What is it?

WIP, throughput and cycle time sat next to each other in the report – never against each other. That comparison is the find in Daniel Vacanti's “Actionable Agile Metrics for Predictability” (2015): divide mean WIP by throughput and Little's Law predicts a mean cycle time. When it is HIGHER than the measured one, some items were finished faster – by borrowing cycle time from others still in progress. Vacanti calls it Flow Debt. Typical causes: expedite lanes, blockers and unspoken pull-order policies.

Why it matters

It is the only view in the tool that makes SILENT PREFERENTIAL TREATMENT visible – without anyone having falsified a number. And it needs no new input: the WIP series is derived from the CFD daily data the pipeline already produces.

How to use it

```
python -m build_reports IssueTimes.xlsx --cfd CFD.xlsx --metrics flow_debt --browser
```

Or in the GUI: the “Flow Debt” checkbox in the metrics list; the “Flow Debt tolerance” field sits below Target CT and is stored in the project template. The CFD is required – without it the metric says so instead of computing.

Three verdicts, one caveat

• Approximation HIGHER than measurement: the process is accumulating Flow Debt – some items are sped up at the expense of others
• Approximation LOWER: the process is paying it off – the items left sitting are coming out now
• Inside the tolerance band: stable. Vacanti gives no number for this third state – without a band the indicator would report Flow Debt almost always. Default 15 %, configurable

Assumptions come before the verdict

Little's Law only holds under conditions. Two of them are checked and stated in the header BEFORE the verdict: arrival and departure rates in balance (assumption 1) and WIP comparable at the start and the end (assumption 3). When they are violated the header says “verdict not dependable” – the number means nothing then, and the report says so instead of letting it stand.